

Image and Video Classification

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Declaration

This report has been prepared on the basis of my own work. Where other published and unpublished source materials have been used, these have been acknowledged.

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Abstract

This comprehensive study delves into the intriguing world of image segmentation and classification using various neural network architectures, from custom models to advanced setups like the improved ResNet-50. Through meticulous experimentation and hyperparameter tuning, we unearth critical insights into how different architectures fare on diverse tasks. The study reveals a fascinating paradox: while advanced models like the improved ResNet-50 excel in complex tasks, achieving a stellar validation accuracy of approximately 97.42%, simpler architectures often outperform them on less intricate datasets. For instance, a custom CNN model achieved a high accuracy of nearly 99.45% on the MNIST dataset, also showcasing remarkable computational efficiency. The research further underscores the pivotal role of hyperparameter tuning in achieving generalizable and efficient models. The findings not only offer a robust foundation for future work but also raise compelling questions about the dynamics between architectural complexity, dataset intricacy, and model performance. This study serves as a valuable guide for both researchers and practitioners looking to navigate the ever-evolving landscape of neural network architectures for image-based tasks.

Table of Content

1	INTRODUCTION	1
1.1	BRIEF OVERVIEW TO IMAGE CLASSIFICATION.....	1
1.2	INTRODUCTION TO DEEP LEARNING.....	1
1.3	AIM AND OBJECTIVES	2
2	LITERATURE REVIEW.....	3
2.1	BACKGROUND RESEARCH ON NEURAL NETWORKS	3
2.1.1	<i>Neural Network</i>	3
2.1.2	<i>Activation Functions:</i>	3
2.1.3	<i>Perceptron</i>	5
2.1.4	<i>Back Propagation Algorithm</i>	5
2.2	MULTILAYER PERCEPTRON.....	6
2.3	CONVOLUTIONAL NEURAL NETWORKS	6
2.4	HYPERPARAMETER TUNING	8
2.5	OVERFITTING AND UNDERFITTING.....	9
3	MODEL BUILDING	10
3.1	DATASET IN ACTION	10
3.2	MODEL BUILDING	11
3.3	IMAGE SEGMENTATION.....	11
3.4	IMAGE CLASSIFICATION ON IRIS (EXPERIMENT 1).....	12
3.5	IMAGE CLASSIFICATION ON DATASET 2:	13
3.6	IMAGE CLASSIFICATION ON MALARIA DATASET	15
4	RESULTS AND ANALYSIS	18
4.1	IMAGE SEGMENTATION.....	18
4.2	IMAGE CLASSIFICATION	20
5	CONCLUSION AND FUTURE WORK.....	23
6	SWOT ANALYSIS	24
7	BIBLIOGRAPHY	26
8	APPENDIX.....	28

1 Introduction

1.1 Brief Overview to Image Classification

In the realm of computer vision, image classification stands as a seminal task, one that has catalysed many technological leaps since its inception. Just as the Rosetta Stone served as a linchpin for deciphering ancient languages, image classification has become a cornerstone for a myriad of applications in our modern landscape. Whether it's facial recognition technology that securely unlocks smartphones or machine learning algorithms that identify disease patterns in medical imagery, the implications of this technology are far-reaching.

Image classification is not just a solitary function; it's a nexus in a network of interconnected tasks. Think of it as the blueprint upon which more complex systems, like object detection and image segmentation, are constructed. Similar to the way that fundamental theorems lay the groundwork in mathematical proofs, image classification provides the foundational logic upon which higher-level vision tasks are developed. In industry and academia alike, the quest for enhanced classification algorithms has inspired a slew of innovations. Researchers are employing a multitude of techniques, from traditional methods like k-Nearest Neighbours to avant-garde approaches involving Convolutional Neural Networks (CNNs). The state-of-the-art models aren't just capstones in academic papers; they are actively deployed in real-world applications such as autonomous vehicles and augmented reality systems.

The technology's transformative potential extends beyond the present. It is poised to unlock opportunities in untapped domains, much like a master key. For instance, effective image classification could revolutionize how we interact with our environment through augmented reality, or optimize how surveillance systems monitor public spaces. The horizon is vast, and as algorithms become more sophisticated, their applicability will only continue to expand. To sum up, image classification serves as both a cornerstone and a catalyst in the ever-evolving domain of computer vision. Its applications are manifold and its potential virtually limitless, echoing the transformative power of the written word in human history. Like the scripts of old that codified the wisdom of ages, this technology has the potential to redefine how we perceive and interact with the world around us.

1.2 Introduction to Deep Learning

Deep learning has emerged as a revolutionary paradigm in the realm of Information Technology over the last half-decade, particularly excelling in predictive analytics. Whereas machine learning serves as a broader umbrella, deep learning operates as its specialized subset, distinguished by architectures that mimic

human neural circuitry. While machine learning continues to be an active research focus, particularly in fields like computer vision and natural language processing, deep learning has expanded the frontier into an array of domains including physics, biochemistry, and earth sciences, among others. The democratization of computational resources, marked by the advent of cloud computing and high-speed internet, has made deep learning models more accessible. Even remote locations can now tap into substantial computational power, thereby widening the scope of what can be achieved in deep learning. One of the quintessential advantages of deep learning lies in its ability for automated feature extraction with minimal user guidance. This autonomy in feature learning has been pivotal in overcoming longstanding challenges like the curse of dimensionality.

Deep learning's academic lineage can be traced back to foundational works on the Perceptron model, which proposed a single-layer neural network. However, computational limitations stymied the field's growth until a watershed moment in 2011. The ImageNet competition, a high-stakes contest in image classification, witnessed a significant performance leap when a neural network model clinched the title. Subsequent innovations have included multi-layered architectures with non-linear activations and polynomial functions, aligning more closely with contemporary deep learning frameworks. Recent advancements have been fueled by algorithmic optimizations and computational breakthroughs. The potent combination of hardware acceleration and increasingly efficient algorithms has propelled deep learning into a dominant position for various complex tasks, setting new benchmarks across multiple domains. With each iteration, models are achieving unprecedented accuracies, solidifying deep learning's position at the forefront of AI research.

1.3 Aim and Objectives

The primary aim of this project is to design, implement, and evaluate deep learning models for image classification and segmentation tasks. The project specifically focuses on optimizing model architectures and hyperparameters to achieve high accuracy and computational efficiency, using both synthetic and real-world datasets. The ultimate goal is to produce models that can be further tested or deployed in larger-scale applications.

Objectives:

1. Literature Review: Conduct a comprehensive review of existing deep learning techniques for image classification and segmentation.
2. Model Development: Create and fine-tune deep learning models, including custom architectures and improved versions of existing models like ResNet-50.
3. Hyperparameter Optimization: Employ grid search or other methods to optimize model hyperparameters for enhanced performance.
4. Performance Evaluation: Assess the models using metrics such as accuracy, loss, and computational time, and compare them to baseline models.
5. Cost-Benefit Analysis: Evaluate the computational and financial costs of the models, aiming to achieve a balance between cost and performance.

2 Literature Review

2.1 Background Research on Neural Networks

2.1.1 Neural Network

Neural networks aim to emulate human brain functions through algorithms, yet they're far from matching the complexity of biological neural systems. These networks employ a series of nonlinear algorithms to find relationships between input variables for predictive outputs. Understanding their notation is crucial for consistent project work. Unlike traditional machine learning, which requires task-specific algorithms, neural networks operate as a 'black box,' making them versatile across various domains. A basic neural network consists of three primary layers: input, hidden, and output. The input layer receives the initial data, its neuron count matching the number of input variables. Hidden layers, which can vary in number, are where the network learns to solve the problem. Fewer hidden layers result in a 'shallow' network, while more layers indicate a 'deep' network. The output layer provides the final prediction. The network's self-guided learning identifies pivotal parameters during the data ingestion process. A neural network's capabilities hinge on three core attributes: expressibility, computational efficiency, and learnability. Expressibility refers to the mathematical framework within which the network's functions operate. Computational efficiency evaluates the balance between resource allocation and the requisite parameters for functional construction. Learnability quantifies the convergence speed of activation functions, which is critical for optimizing the network's performance.

2.1.2 Activation Functions:

The activation function governs the neuron's responsiveness within a neural network, serving as a nonlinear gatekeeper that enables the model to capture intricate data patterns. By determining neuron activation based on input, it introduces the necessary non-linearity for complex learning. For backpropagation to operate smoothly, activation functions should be differentiable and continuous. In this project, we employ a range of rigorously-tested activation functions—Sigmoid, Tanh, ReLU, Linear, Gauss, and SoftMax—to optimize network performance. While many nonlinear functions exist, only a select few have proven effective as activation functions through empirical validation.

Sigmoid is widely employed in feed-forward neural networks due to its non-linear properties, enabling the network to effectively capture complex data patterns. Being a continuous and differentiable function, it facilitates efficient backpropagation by providing real-valued derivatives. The activation function for sigmoid is represented as:

$$\sigma(s_k) = \frac{1}{1 + e^{-s_k}} = \frac{e^{s_k}}{1 + e^{s_k}}$$

Tanh serves as an effective non-linear activation function operating near zero, resembling a step function in its behavior. Due to its non-linearity, Tanh enables neurons to capture intricate data patterns. Being both continuous and differentiable, Tanh facilitates smooth operation of backpropagation algorithms. Compared to sigmoid, neural networks generally exhibit enhanced performance when utilizing Tanh as the activation function. Tanh is mathematically represented by:

$$\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$$

In recent years, ReLU (Rectified Linear Unit) has become the activation function of choice, outperforming alternatives like Sigmoid and Tanh across diverse datasets. Unlike its highly nonlinear counterparts, ReLU possesses a near-linear graph, simplifying the gradient descent optimization process. As a threshold-based function, ReLU has set the standard for activation functions globally. The mathematical definition of ReLU is as follows:

$$\text{ReLU}(x) = \max(0, x)$$

The Linear activation function is often equated to having no activation function due to its minimal impact on neural network performance. Technically, a single-layer network with a Linear activation function is considered a perceptron. While it has merits, its primary drawback is a constant derivative of zero, which impedes weight updates during backpropagation and thus stifles learning. In cases where the input features are continuous, a single-layer network with this function essentially performs Linear regression. The activation function is formally denoted by:

$$f(x) = a \cdot x$$

The Gaussian activation function is a nonlinear activation function with the gaussian surface being used as the gaussian activation function. The function is continuous, and the derivative is continuous makes the function to build a network so that weight gets updated. This doesn't perform as well as ReLU or sigmoid or Tanh but performs better than Linear Activation function. The activation function is represented by

$$f(x) = e^{-x^2}$$

SoftMax activation function is a non-linear activation function, which creates The SoftMax activation function introduces non-linearity into the neural network, enabling the model to learn complex parameter relationships. It produces output in the form of a probability distribution. The function is mathematically represented by:

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

Forward Propagation

Forward propagation is a process in which at every hidden layer, the output is calculated, and it is propagated to the next layer and the output is calculated at every node, this propagation of activation of neuron at every node in a forward path till it reaches the output is called Forward propagation. In short the process of Input travelling from input layer to the hidden layers and reaching the output layers is called Forward Propagation.

Back Propagation

Backpropagation is critical for neural network training, tasked with weight adjustments through gradient-based optimization. The algorithm computes output discrepancies to iteratively refine weights, often employing the Adam optimizer for enhanced efficacy.

Feed forward neural network is the basic of any neural network, as the structure is very easy and straight forward. In this network, the output goes from input layer to the output layer with no hidden layers in it, the reason they are called feed forward is because from input to output the information transfer is only one way, there is no backpropagation.

2.1.3 Perceptron

Perceptron [1] is probably the first neural network, with a wide range of applications. In 1958 it didn't as much prominence as now due to very little computational power at that time. The activation function used in the perceptron is Identity function, which is also the same for Linear regression. In terms of mathematical approach, they both act a like. The below figure represents the perceptron network. Although it was there since 1958, perceptron didn't again much interest among the researchers as it couldn't classify various pattern recognition problems. Perceptron works efficiently when the data set is linearly separable.

2.1.4 Back Propagation Algorithm

Back propagation Algorithm [3] is one of the oldest algorithms in the field of deep learning and it gave a new perspective to the approach of learning algorithms in general. This is a technique used in modern deep learning to update the weights of the neurons. This algorithm allows us to use the chain rule of derivatives of functions that are used in the computation of network. There are many different ways in which this process can be optimized. There are so many optimizers defined in order to optimize the back propagation algorithm. Adam is used extensively for the past 5 years across industries and researchers.

One Hot encoding

One hot encoding is a trick that is by converting the complicated categorical variables into simple ones and zeros. The higher the number of categorical variables, larger the size of the matrix.

Data Normalisation Data normalization has emerged as a critical modeling component in the current landscape, primarily due to its power in mitigating the influence of outliers on models. The core objective of normalization is to establish equitable significance across variables, regardless of the presence of outliers within the

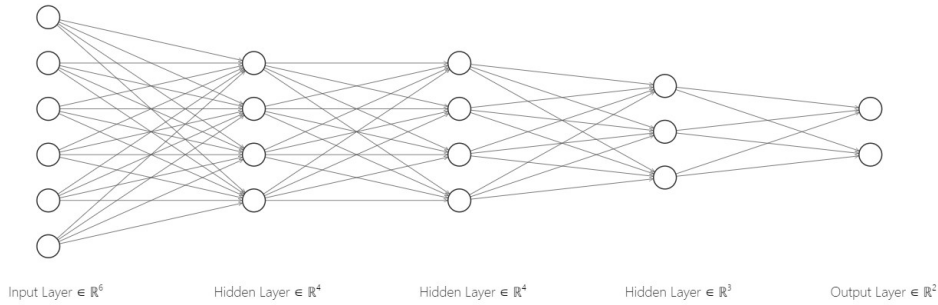
dataset. Multiple techniques have been devised and extensively employed in the realm of machine learning, with z-score normalization, Rescaling, and Mean Normalization standing out as popular options.

Among these techniques, Rescaling, also known as Min-Max normalization, holds a prominent position due to its simplicity and scalability. It involves transforming variables into a functional range, ensuring consistency in their impact on the model. Mean Normalization and standardization share a similar rationale for data normalization. The primary distinction lies in the denominator used for scaling. Mean Normalization employs the range between the maximum and minimum values, while standardization utilizes the standard deviation. Standardization is colloquially known as z-score transformation.

In essence, these normalization methods collectively contribute to refining data preprocessing in machine learning, bolstering model performance and resilience against outliers.

2.2 Multilayer Perceptron

The multi-layer perceptron (MLP) is an advanced version of the perceptron network achieved by introducing additional layers. Its effectiveness stems from its increased layer count, which enhances the network's capability to discern multiple data patterns. While the perceptron originally utilized an identity activation function, MLP employs diverse activation functions, introducing crucial non-linearity that facilitates more efficient data pattern learning. The output calculation involves the weighted sum of the activation function and the prior network state, contributing to MLP's comprehensive learning capacity.



Custom Chart : MLP drawn (using tool)

2.3 Convolutional Neural Networks

Convolutional Neural Networks (CNNs), specialized for image processing tasks, surpass Fully Connected Multi-Layer Perceptron's (FC-MLPs) in computational efficiency. This efficiency stems from the convolution operation, a technique borrowed from signal processing. The landmark CNN model, LeNet, was developed at AT&T Bell Laboratories and was instrumental in showcasing the power of the backpropagation algorithm on a global scale. It successfully classified

handwritten images, paving the way for CNNs. Both CNNs and FC-MLPs operate on similar principles involving neurons equipped with weights, biases, and activation functions. However, the architecture of CNNs enables greater efficiency, particularly in handling image data. A typical CNN architecture comprises three core components: Convolutional Layers, Pooling Layers, and Fully Connected Layers. These elements collaboratively reduce computational complexity and the total number of required neurons, thereby making CNNs especially well-suited for image analysis tasks.

For effective model development, one must grasp the quantity of trainable parameters. Given input dimensions ($m \times m$) and output dimensions $n \times n$, the number of output dimensions can be determined using the kernel size k , padding p and stride s in various configurations

Without padding or stride

$$n = m - k + 1$$

With padding and no stride

$$n = m + 2p + 1$$

When employing a stride s without padding,

$$n = \frac{m}{s} + 1$$

With padding p , kernel size k , stride size s

$$n = \frac{m + 2p - k}{s} + 1$$

Total Number of Trainable Parameters

$$P = (k * k * c + 1) * n$$

Forward Propagation in CNN

Forward Propagation in CNN works mostly similar to the traditional Neural network, but function of convolution makes it special, decreases the number of neurons required to compute the same function with computational efficiency. The convolutional layer extracts the features from the input image and the fully connected dense network makes uses the convolution layer to generate the output in the Forward propagation of CNN.

Back Propagation in CNN

Back Propagation in CNNs parallels its counterpart in Multi-Layer Perceptrons (MLPs) with some distinct features. Specifically, the convolution operation is not limited to the Forward Propagation phase; it is also instrumental during Back Propagation. Here, not only are the weights updated, but the convolutional filters themselves are fine-tuned. This dual update mechanism enhances the network's ability to recognize and adapt to intricate patterns, thus elevating its computational efficacy.

2.4 Hyperparameter Tuning

Hyperparameters steer the learning behaviour of deep learning models, affecting areas such as parameter optimization and feature selection. These are generally classified into two major types: architecture-specific hyperparameters and training-process hyperparameters. The first category shapes the structure of the deep learning model, specifying elements like layer depth, the number of neurons per layer, filter dimensions, stride sizes, padding, dropout rates, and the types of activation functions. The second category influences the mechanics of the training phase, affecting update sizes during backpropagation through variables like learning rate and momentum, as well as the choice of optimization techniques and specialized loss functions. The process of training a deep learning model comes with the inherent risks of either overfitting or underfitting. While underfitting is usually due to an imbalance of excessive parameters against limited data, overfitting occurs in the opposite scenario. Although minor overfitting is often tolerable in deep learning, fine-tuning hyperparameters to achieve a harmonious balance between training and validation outcomes is recommended. This ensures a more generalizable model, achievable via regularization methods like weight decay or L1/L2 regularization. The specific hyperparameters employed in this project will be discussed subsequently.

Architecture-Specific Hyperparameters

These hyperparameters define the blueprint of the deep learning model, setting the foundational components for the neural network's architecture.

- **Layer Depth:** Specifies the number of layers in the neural network, determining its complexity.
- **Neurons Per Layer:** Dictates the number of neurons in each layer, which in turn influences the model's capacity to capture various features.
- **Filter Dimensions:** These are the dimensions of the filters used in convolutional layers, affecting how local features are captured.
- **Stride Sizes:** These set how much the filter moves during each convolution operation, influencing the granularity of feature extraction.
- **Padding:** Decides whether zeros should be added around the input in convolutional layers, affecting the output dimensions.
- **Dropout Rates:** Defines the rate at which individual neurons are randomly turned off during training, aiding in regularization.
- **Activation Functions:** Nonlinear functions like ReLU or sigmoid applied to the output of each neuron, determining the complexity of functions that the model can learn.

Training-Process Hyperparameters

These hyperparameters manage the model's learning dynamics, influencing the optimization of trainable parameters.

- **Learning Rate:** Sets the step size during the optimization process, influencing how quickly or slowly a model learns.

- **Optimization Techniques:** Algorithms like SGD, Adam, or RMSprop used to minimize or maximize performance metrics.
- **Loss Function:** The loss function plays a critical role in optimizing neural network learning. Essentially quantifying the discrepancy between predicted and actual values, its minimization aims to improve overall model performance. In this project, we'll employ categorical loss as our chosen loss function.
- **Batch size:** Batch size dictates the number of training samples used in each iteration of model training. For instance, if a dataset contains 10,000 samples and the batch size is set to 100, the model will update its weights 100 times per epoch, each time using a different set of 100 samples. This approach avoids loading the entire dataset into memory, thereby optimizing disk usage and computational efficiency. Choosing an appropriate batch size is crucial for model performance and training speed. Larger batch sizes offer more accurate gradient estimates but at the cost of increased memory usage, while smaller sizes are computationally efficient but may lead to noisy gradient updates.

2.5 Overfitting and Underfitting

Overfitting is a pervasive issue in deep learning where models memorize data rather than generalizing from it. A quick diagnostic tool is to plot training versus validation accuracy; a significant gap indicates overfitting. Conversely, underfitting happens when models can't capture data patterns due to insufficient training data. Both scenarios reflect an inability to generalize: overfitting memorizes data, while underfitting lacks the data needed for pattern recognition.

3 Model Building

3.1 Dataset in Action

In this project as a part of implementation, I will be choosing two datasets for implementation. The reason for choosing two datasets is that to get the familiarity with the process, I am trying to use MNIST which is one of most widely used dataset for training and academic purposes. The next dataset is a Malaria parasite detection.

MNIST Dataset

The MNIST (Modified National Institute of Standards and Technology) dataset serves as a foundational resource in the machine learning community for tasks involving digit recognition. Comprising 60,000 training images and 10,000 test images, the dataset includes handwritten digits ranging from 0 to 9, making for 10 distinct classes. Each grayscale image has dimensions 28x28, resulting in a 784-pixel vector representation. This choice of image size facilitates easier computational processing. The dataset is freely available for use and is often accessed via TensorFlow for Convolutional Neural Networks (CNN) implementations.

Regarding data layout:

- 'X_train': 60,000 samples, each of size 28x28
- 'X_test': 10,000 samples, also of size 28x28
- 'Y_train': 60,000 labels corresponding to X_train
- 'Y_test': 10,000 labels corresponding to X_test

It's worth noting that the MNIST dataset is actually a subset of a larger, more comprehensive dataset known as NIST. Additionally, the dataset is typically provided in CSV format, where each cell represents the intensity level of a corresponding pixel.

TensorFlow Malaria Dataset

The TensorFlow Malaria dataset is a specialized resource aimed at facilitating research in detecting malaria parasites in blood smear images. It contains a large number of microscopic images labeled either as 'Parasitized' or 'Uninfected,' making it a binary classification problem. The images vary in size and are generally of high resolution to capture the nuanced features necessary for accurate classification. This dataset is widely utilized in the development of CNNs, often serving as a benchmark for medical image analysis algorithms.

For a more technical layout:

- 'X_train': A number of high-resolution samples (the exact count varies by version)
- 'X_test': A separate set of high-resolution samples for evaluation
- 'Y_train': Labels for the training set, either 'Parasitized' or 'Uninfected'
- 'Y_test': Labels for the test set, also either 'Parasitized' or 'Uninfected'

The dataset is available through TensorFlow's data API, making it accessible for rapid model development and evaluation. Both of these datasets are rich in features and applications, contributing to their sustained relevance in the machine learning community.

3.2 Model Building

There are several models that will be implemented as a part of this project. In the initial phase I will be building the model with K-NN and SVM. The next step is to build a model entirely from the scratch and implement it on top of MNIST and Greek character recognition. In the phase 2 I will be using TensorFlow to build the CNN model. I will be varying all the possible Hyper parameters and find the best optimized solution for this problem. The model I will be building are as follows

3.3 Image Segmentation

Experiment 1: Creating a Custom Model for Image Segmentation

In the first experiment, the aim is to create a custom model for image segmentation tasks. The data used for this experiment is synthetic, generated to have 128x128 images with randomly positioned circles serving as masks. The model architecture designed is a simple ConvNet, composed of Conv2D layers for feature extraction, MaxPooling layers for down-sampling, and Conv2DTranspose layers for up-sampling. The model uses a sigmoid activation function in the output layer to generate binary masks. After defining the model, it is compiled using the Adam optimizer and binary cross-entropy loss function. The model is then trained on the synthetic data with a batch size of 32 for 10 epochs. After training, predictions are made on a subset of 5 images, and these predictions are visualized alongside the true masks and original images to assess the model's performance.

Experiment 2: Generating Synthetic Data (Duplicate)

The second experiment appears to be a duplicate of the data generation step from Experiment 1. Similar to the first experiment, synthetic data is generated with circles as masks in 128x128 images. No further steps are undertaken, making it an isolated data generation step.

Experiment 3: Building Upon Baseline MobileNetV2

The third experiment takes a different approach by building upon a pre-trained MobileNetV2 model. The synthetic data generated in Experiment 2 is used here. The model architecture is extended by adding Conv2D layers for additional feature extraction and Conv2DTranspose layers for up-sampling. This combines the power of pre-trained features with custom layers tailored for segmentation. After defining the architecture, the model is compiled and trained in the same manner as in Experiment 1, using Adam optimizer and binary cross-entropy loss. Post-training, predictions are made for 5 images, which are then visualized to assess how well the model performs in comparison to the first experiment.

Experiment 4: Image Segmentation for Oxford Dataset

The fourth experiment focuses on segmenting images from the Oxford-IIIT Pet dataset. Initially, the dataset is loaded and preprocessed to resize and normalize the images and masks. A custom Keras layer for data augmentation is also introduced, implementing random horizontal flipping. The model architecture used

is a U-Net model, built upon a pre-trained MobileNetV2 base. The U-Net architecture includes down-sampling from the base model and up-sampling using layers from the pix2pix example. The model is compiled using Adam optimizer and Sparse Categorical Cross-Entropy loss, given that there are multiple classes in the output masks. Training is conducted for 20 epochs, and the training and validation loss are plotted for evaluation. Finally, the model's predictions are visualized for a subset of the test dataset, showcasing how well the model performs on real-world data.

3.4 Image Classification on IRIS (experiment 1)

This is an experimental dataset, and an extremely small and popular dataset, IRIS dataset. Experiment: Classification of the Iris Dataset Using Neural Networks
Data Preparation

First, the Iris dataset is loaded using scikit-learn's `load_iris` function, which provides the feature matrix `X` and target labels `y`. The dataset is then split into training and testing sets, allocating 80% of the data for training and 20% for testing. The random seed is set to 42 for reproducibility.

Data Preprocessing

After splitting, the features are normalized by dividing them by the maximum value in each feature column, effectively scaling them between 0 and 1. This is done to make the training process more stable and faster, as the optimizer can work more efficiently with normalized data. The labels are one-hot encoded, turning them into a format suitable for multi-class classification. Specifically, the labels are converted into vectors of length 3 (since there are three classes), where the index corresponding to the class label is set to 1, and all other indices are set to 0.

Model Architecture

A neural network is then constructed using TensorFlow's Keras API. The model consists of three layers:

- The first layer is a dense layer with 64 neurons and a ReLU activation function. It also specifies the input shape as 4, corresponding to the four features in the Iris dataset.
- The second layer is a dense layer with 32 neurons and a ReLU activation function.
- The third layer is the output layer, a dense layer with 3 neurons (one for each class) and a softmax activation function to produce probabilities for each class.

Model Training and Evaluation

The model is compiled using the Adam optimizer and categorical cross-entropy loss, suitable for multi-class classification problems. The metric used for evaluation is accuracy. The model is trained for 50 epochs using the training data, and validation is performed using the testing data. The training process returns a history object that captures the training and validation metrics over time. The Test accuracy obtained is 96.67%.

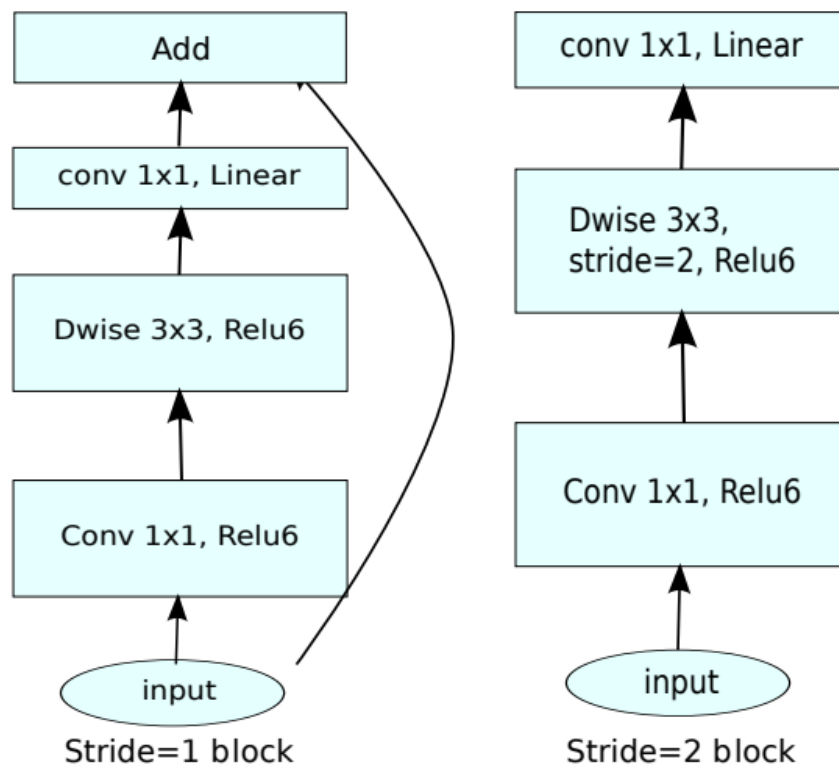
3.5 Image Classification on Dataset 2:

Data Preprocessing

Before any experiment, the MNIST dataset is loaded, and the data is pre-processed. The images are reshaped to a 4D array suitable for CNN models and normalized by dividing by 255. The labels are one-hot encoded to make them compatible for classification tasks. These preprocessing steps are essential for the efficient training of neural networks.

Experiment 1: MobileNetV2 Model

In this experiment, the MobileNetV2 architecture is utilized, which is a model known for its efficiency on mobile devices. The architecture is loaded with pre-trained weights from the ImageNet dataset but excludes the top layer. Custom layers are then added to tailor the model for the MNIST dataset. Specifically, a Global Average Pooling 2D layer is added to reduce dimensionality, followed by a Dense layer with 1024 units and ReLU activation, and finally another Dense layer for the 10-class output with softmax activation. The MNIST images are resized to 96x96 pixels and converted to RGB to match MobileNetV2's input requirements. The model is then compiled and trained for 5 epochs.



Experiment 2: Custom CNN Model 1

This experiment focuses on a custom-built Convolutional Neural Network

Model: "model"

Layer (type)	Output Shape	Param #
input_1 (InputLayer)	[(None, 28, 28, 1)]	0
conv2d (Conv2D)	(None, 28, 28, 64)	640
max_pooling2d (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_1 (Conv2D)	(None, 14, 14, 128)	73856
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 128)	0
flatten (Flatten)	(None, 6272)	0
dense (Dense)	(None, 256)	1605888
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 10)	2570

=====
Total params: 1,682,954
Trainable params: 1,682,954
Non-trainable params: 0

(CNN) tailored for image classification tasks like MNIST. The model consists of two blocks: each block contains a Conv2D layer followed by a MaxPooling2D layer. The Conv2D layers are responsible for feature extraction, and MaxPooling2D layers reduce dimensionality. After these blocks, the data is flattened, and two Dense layers are used for classification. The model uses dropout for regularization. It's a simpler model compared to MobileNetV2 but well-suited for a task like MNIST classification. The model is compiled, trained for 5 epochs, and then evaluated on the test set. The training and validation accuracies are plotted for analysis.

Experiment 3: Complex CNN Model 2

This experiment involves a more complex CNN model with additional techniques like Batch Normalization and L1/L2 regularization. It contains three blocks of Conv2D and MaxPooling2D layers, with each block followed by Batch Normalization and Dropout layers. The Batch Normalization layers help in faster and more stable training, while the Dropout and regularization techniques aim to prevent overfitting. This model also has more layers and more filters in the Conv2D layers, making it more complex and potentially more accurate but at the cost of higher computational needs. After training for 5 epochs with a batch size of 128, the model is evaluated, and its performance metrics are plotted. Each experiment serves a different purpose: the first leverages a pre-trained model for quick and potentially powerful results; the second uses a simple but effective custom CNN tailored for the task; and the third employs a more complex architecture with added regularization and normalization techniques. Each approach has its own trade-offs in terms of accuracy, computational cost, and complexity.

```

Model: "model_2"
Layer (type)                 Output Shape                 Param #
-----
input_3 (InputLayer)         [(None, 28, 28, 1)]         0
conv2d_5 (Conv2D)            (None, 28, 28, 64)          640
batch_normalization_3 (Batch Normalization) (None, 28, 28, 64)          256
max_pooling2d_5 (MaxPooling2D) (None, 14, 14, 64)          0
dropout_5 (Dropout)          (None, 14, 14, 64)          0
conv2d_6 (Conv2D)            (None, 14, 14, 128)         73856
batch_normalization_4 (Batch Normalization) (None, 14, 14, 128)         512
max_pooling2d_6 (MaxPooling2D) (None, 7, 7, 128)          0
dropout_6 (Dropout)          (None, 7, 7, 128)          0
conv2d_7 (Conv2D)            (None, 7, 7, 256)           295168
batch_normalization_5 (Batch Normalization) (None, 7, 7, 256)           1024
max_pooling2d_7 (MaxPooling2D) (None, 3, 3, 256)          0
dropout_7 (Dropout)          (None, 3, 3, 256)          0
flatten_2 (Flatten)          (None, 2304)                 0
dense_4 (Dense)              (None, 512)                  1180160
dropout_8 (Dropout)          (None, 512)                  0
dense_5 (Dense)              (None, 10)                   5130
-----
Total params: 1,556,746
Trainable params: 1,555,850
Non-trainable params: 896

```

3.6 Image classification on Malaria Dataset

Custom CNN Model

Data Preprocessing

The code starts with data preprocessing steps where the Malaria dataset is loaded from TensorFlow Datasets. Each image is resized to dimensions of 70x70 and normalized by dividing the pixel values by 255. These steps are essential to make the data compatible with CNN models and to speed up training. The dataset is then split into training (80%) and validation (20%) sets. Finally, the datasets are batched, with a batch size of 64, and the training set is shuffled to ensure randomization.

Hyperparameter Space Definition

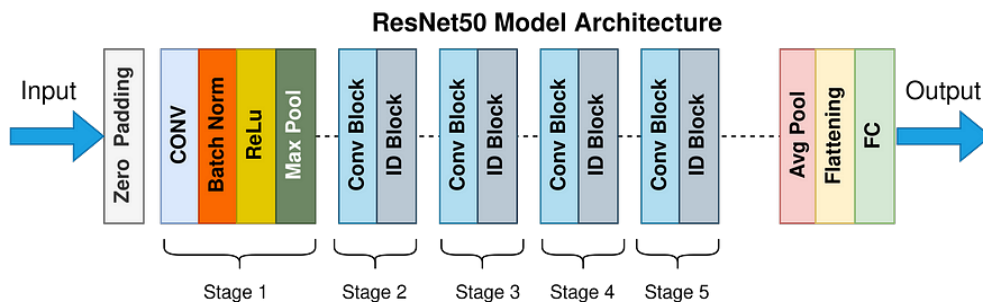
Before diving into the hyperparameter tuning experiment, the code specifies the hyperparameter space. This is an important step because it defines the

boundaries within which the model will be optimized. The hyperparameters under consideration are optimizer types (Adam and SGD), dropout rates (0.2, 0.3, 0.4), units in the Dense layer (128, 256, 512), learning rates (0.001, 0.01), and activation functions (ReLU, Tanh).

Hyperparameter Tuning Experiment

The core of the code lies in the hyperparameter tuning experiment. A nested loop iterates through every possible combination of hyperparameters to train a CNN model. For each combination, a model is instantiated with the specified hyperparameters. The CNN model consists of two Conv2D layers for feature extraction, MaxPooling layers for dimensionality reduction, and Dense layers for classification. The model is then compiled and trained on the training set for 5 epochs. After each training cycle, the model's performance is evaluated on the validation set, and the results are stored. If the current model's validation accuracy surpasses the best observed so far, the best score and corresponding parameters are updated.

Original ResNET-50



Improved RESNET-50

Architecture:

Improved ResNet-50 Architecture

- Initial Conv Layer: Begin with a 7x7 convolutional layer with 64 filters, followed by Batch Normalization and ReLU activation.
- First Set of Residual Blocks:
 - Use 3 residual blocks with 64 filters.
 - For the first block, use a convolutional shortcut to match dimensions.
- Second Set of Residual Blocks:
 - Use 4 residual blocks with 128 filters.
 - For the first block, increase the stride to 2 for downsampling and use a convolutional shortcut.

- Third Set of Residual Blocks:
 - Use 6 residual blocks with 256 filters.
 - Again, for the first block, increase the stride to 2 and use a convolutional shortcut.
- Fourth Set of Residual Blocks:
 - Use 3 residual blocks with 512 filters.
 - For the first block, increase the stride to 2 and use a convolutional shortcut.
 - Global Average Pooling: Use a Global Average Pooling layer to reduce dimensions.

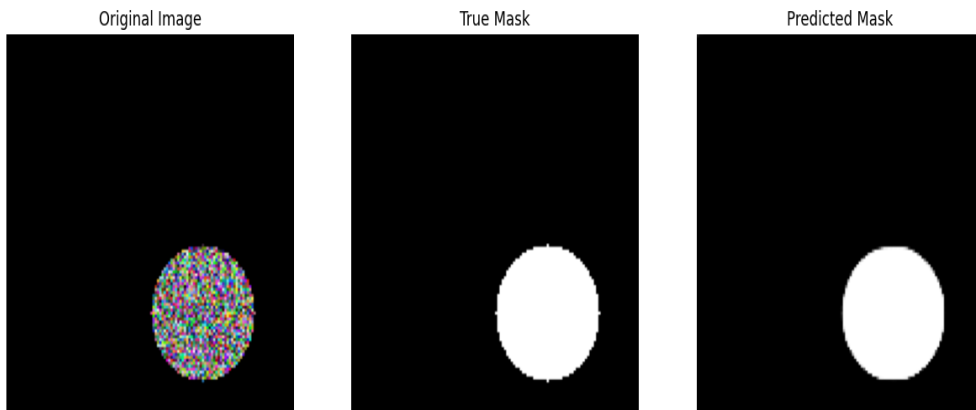
Fully Connected Layer: Finally, add a fully connected layer with 1 unit with a sigmoid activation for binary classification. Same kind of Hyper-parameter tuning has been performed.

4 Results and Analysis

4.1 Image Segmentation

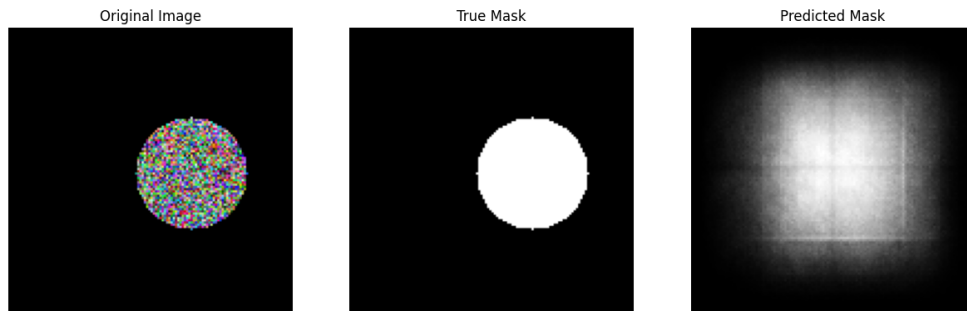
Experiment 1: Creating a Custom Model for Image Segmentation

In Experiment 1, the loss metric shows a strong performance from the model. The training loss starts at 0.4606 in the first epoch and rapidly declines to 0.0024 by the 10th epoch. The validation loss also decreases in a similar fashion, starting at 0.0143 and reaching 0.0027. The rapid decline in both training and validation loss indicates that the model quickly adapted to the synthetic data, achieving low loss values. This suggests effective feature extraction and segmentation capabilities. However, one point to note is that the synthetic data might be too simple to gauge the model's effectiveness for more complex real-world tasks.



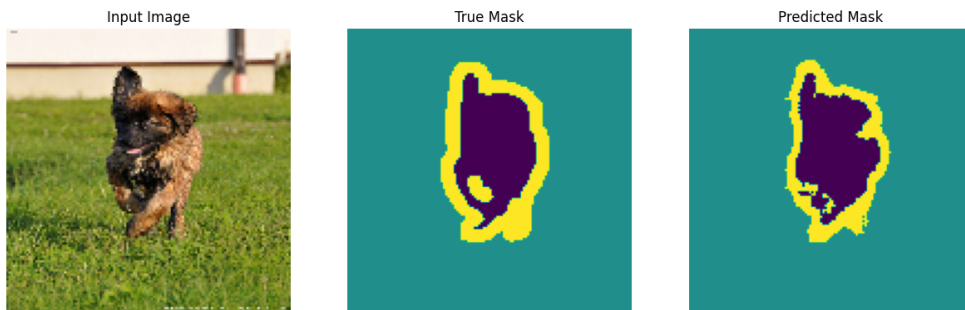
Experiment 3: Building Upon Baseline MobileNetV2

For Experiment 3, the training and validation loss metrics are higher compared to Experiment 1. The training loss starts at 0.4035 and marginally improves to 0.2636 by the end of the 10th epoch. The validation loss starts at 0.2688 and slightly decreases to 0.2625. The minor change in loss values suggests that the model might be struggling to adapt to the synthetic dataset. This could be due to the complexity added by the MobileNetV2 architecture, making it less suited for the synthetic data. It's also possible that the model is not fully optimized, given the relatively stable loss metrics throughout the epochs.



Experiment 4: Image Segmentation for Oxford Dataset

In Experiment 4, both loss and accuracy are considered. The training loss starts at 0.5107 and consistently reduces to 0.1757 over 20 epochs. The training accuracy starts at 80.18% and improves to 92.82%. Validation loss starts at 0.4332 and shows a more varied pattern, reaching 0.2802 by the 20th epoch. Validation accuracy starts at 85.35% and improves to 90.11%. While the model shows consistent improvement in training metrics, the validation metrics fluctuate, especially the loss. This may indicate the model is starting to overfit, as it performs well on training data but less consistently on validation data. The accuracy of around 90% on a real-world dataset like Oxford-IIIT Pet also indicates a fairly strong model.



Conclusions

- Experiment 1 showed rapid learning and low loss, but the synthetic dataset might not be complex enough to validate the model's robustness.
- Experiment 3 had relatively stable loss metrics, suggesting it might not be as well-optimized for the synthetic dataset as the custom model in Experiment 1.
- Experiment 4 displayed consistent improvement in training metrics but fluctuating validation metrics, suggesting the onset of overfitting. Despite this, achieving around 90% accuracy on a real-world dataset indicates good performance.

4.2 Image Classification

Experiment 1: MobileNetV2 Model

The MobileNetV2 model shows suboptimal performance on the MNIST dataset, starting with an accuracy of 35.37% on the training set and 27.44% on the validation set. By the fifth epoch, the accuracy increases only modestly to 41.23% on the training set and 40.82% on the validation set. Given that MobileNetV2 is a powerful architecture designed for more complex tasks, this suggests that the model is not well-suited for the MNIST dataset. The high computational time (approximately 1555 seconds for the fifth epoch) also makes it inefficient. The validation loss also fluctuates, indicating that the model struggles to generalize well for this task.

Experiment 2: Custom CNN Model 1

This experiment shows drastically better performance than the MobileNetV2 model. The training starts with an accuracy of 96.25% and a validation accuracy of 98.70% in the first epoch. By the fifth epoch, the model achieves 99.45% accuracy on the training set and 99.23% on the validation set. The model is both computationally efficient (around 135 seconds for the fifth epoch) and highly accurate, making it well-suited for the MNIST dataset. The losses also decrease steadily, confirming that the model is learning effectively and generalizing well.

Experiment 3: Complex CNN Model 2

In this experiment, the complex CNN model performs exceedingly well but with a few caveats. Starting with a training accuracy of 92.08% and a validation accuracy of 76.94%, the model improves significantly by the second epoch, achieving nearly 98.85% validation accuracy. However, the computational time is relatively high (around 246 seconds for the fifth epoch). Additionally, the model's performance appears to plateau by the fourth epoch, with the validation accuracy hovering around 99.14%. The fifth epoch shows a slight decrease in validation accuracy to 98.75%, suggesting that the model might be starting to overfit, despite the regularization techniques employed.

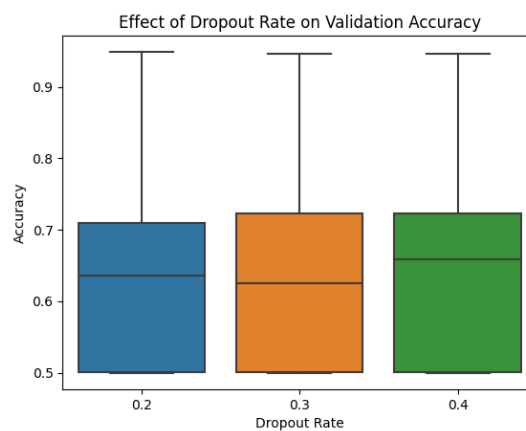
Conclusions

- MobileNetV2: Unsuitable for MNIST, both in terms of accuracy and computational efficiency.
- Custom CNN Model 1: Highly efficient and accurate, making it ideal for simple image classification tasks like MNIST.
- Complex CNN Model 2: Achieves high accuracy but at a higher computational cost and might be prone to overfitting in the long run.
- The results indicate that for a dataset like MNIST, simpler architectures can often yield better performance both in terms of accuracy and computational time.

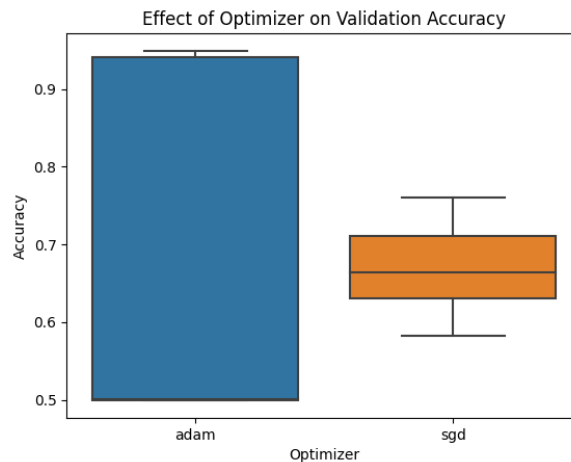
Image Classification on Malaria dataset

Custom Made Model

The best model achieved a validation accuracy of approximately 94.9%, which is a strong performance for a binary classification task on the Malaria dataset. Let's dissect the best hyperparameters and their implications: The Adam optimizer was identified as the best choice, which is consistent with its reputation for fast and efficient optimization in a wide variety of machine learning tasks. It dynamically adjusts the learning rate during training, which likely contributed to the high accuracy. A dropout rate of 0.2 was optimal, indicating that a small amount of regularization was beneficial for the model. This suggests that the model was complex enough to fit the data well but not so complex as to overfit, given that a low dropout rate sufficed for regularization. The optimal number of units in the Dense layer was found to be 512. This suggests that a higher capacity in the fully connected layer was beneficial for capturing the complexity of the data. Given the high-dimensional input (70x70 images), this makes sense as more units can capture more features for classification. A learning rate of 0.001 was found to be optimal, which is a standard choice for many optimization algorithms. This indicates that the model was able to learn effectively at this rate without overshooting the minimum loss or learning too slowly. The ReLU (Rectified Linear Unit) activation function was found to be the most effective among the choices. This is consistent with its widespread use in deep learning due to its advantages such as computational efficiency and sparsity.



In summary, the hyperparameter tuning experiment successfully identified a strong combination of settings that achieved a high validation accuracy. The results are aligned with conventional wisdom about the effective use of Adam optimizer, ReLU activation, and moderate regularization. The high number of units in the Dense layer further contributed to the model's ability to capture complex patterns in the data. Overall, the experiment appears to have been successful in optimizing the model for this specific task.



Improved ResNet50

Best Score: 0.9742420563355917

Best Params: {'optimizer': 'adam', 'dropout_rate': 0.2, 'units': 512, 'learning_rate': 0.001, 'activation': 'relu'}

The improved ResNet-50 model achieved a stellar validation accuracy of approximately 97.42%, a considerable improvement over the previous architecture's 94.9%. This underlines the effectiveness of the improvements made to the original ResNet-50 architecture. Let's break down the selected hyperparameters: Once again, the Adam optimizer proved to be the most effective, corroborating its reputation for speedy and efficient convergence. It's especially well-suited for deep networks like ResNet, which have a large number of layers. A dropout rate of 0.2 was optimal, just like in the previous experiment. This suggests that the model was able to generalize well without needing heavy regularization. Given the complexity and depth of ResNet, a low dropout rate seems sufficient. The model preferred a dense layer with 512 units, implying that the capacity at the tail end of the network is crucial for making accurate final predictions. This makes sense as ResNet architectures are known to capture highly abstract features, and a sufficiently large dense layer can make the most of these. A learning rate of 0.001 was selected, indicating that the network was able to learn efficiently without overshooting or getting stuck in local minima. This is a commonly used learning rate and often works well for deep networks. ReLU was chosen as the activation function, which is in line with general best practices for deep learning models. Given its non-saturating nature, ReLU is often better for training deeper networks. In summary, the hyperparameter tuning process has once again identified an effective combination of settings, but this time for a much deeper and more complex architecture. The result is a model that performs exceptionally well on the validation set, and the choice of hyperparameters indicates a well-generalized model. The high accuracy rate makes this tuned, improved ResNet-50 model a strong candidate for further testing or deployment.

5 Conclusion and Future Work

In conclusion, the series of experiments conducted across different domains, from image segmentation to image classification, have provided valuable insights into the capabilities and limitations of various architectures. Among these, the improved ResNet-50 architecture stood out, achieving a high validation accuracy of approximately 97.42% and indicating its suitability for more complex, real-world applications. This performance was achieved through meticulous hyperparameter tuning, underscoring the importance of this step in building generalizable models. On the other hand, the image segmentation experiments offered a nuanced view. While the custom model in Experiment 1 showed rapid learning and low loss metrics, the synthetic nature of the data used necessitates further validation on more complex datasets. Experiment 3 with MobileNetV2 indicated that not all advanced architectures are well-suited for every type of data, as evidenced by its stable but unimpressive loss metrics. Experiment 4, although achieving high accuracy, raised concerns about overfitting, a recurring theme also observed in Experiment 3 of the image classification tasks. Specifically, in the image classification domain, the experiments highlighted an interesting observation: simpler architectures could be more effective for simpler datasets. This was evident in Experiment 2, where the custom CNN model not only achieved high accuracy but also demonstrated superior computational efficiency compared to its more complex counterparts. In contrast, the complex CNN model in Experiment 3, although highly accurate, showed signs of overfitting, despite the incorporation of regularization techniques. Overall, the results from these experiments offer a robust foundation for future work, emphasizing that the choice of architecture and hyperparameters can significantly impact a model's performance and efficiency.

In terms of future scope, several avenues could be explored to extend the work conducted in these experiments. One promising direction is the incorporation of advanced data augmentation techniques, which could bolster the model's ability to generalize and adapt to more complex datasets. The compelling performance of the improved ResNet-50 architecture also opens up the possibility of employing other pre-trained models, particularly for complex tasks in both image segmentation and classification. Additionally, while the Adam optimizer consistently outperformed other algorithms, exploring alternative optimization techniques such as AdaGrad or RMSProp may provide incremental gains. Real-world testing remains a critical next step; the models should be evaluated on more varied and complex datasets to validate their efficacy fully. Attention could also be directed towards optimizing computational efficiency, especially given the growing emphasis on deploying machine learning models in resource-constrained environments. Given the increasing importance of explainability in AI, integrating layers or methods for model interpretability could also be beneficial, particularly for applications in critical domains like healthcare. For specific tasks such as the classification of pet breeds in the Oxford-IIIT dataset, the implementation of fine-grained classification techniques could offer further improvements. Lastly, although significant effort was put into hyperparameter tuning, more advanced techniques like Bayesian Optimization could yield even more finely-tuned models.

6 Swot Analysis

Strengths:

- **Comprehensive Model Evaluation:** The project involved thorough testing of various neural architectures, providing a holistic understanding of their performance across different types of data.
- **Hyperparameter Tuning:** The inclusion of hyperparameter optimization strengthened the models and gave insight into their robustness and generalizability.
- **Academic Guidance:** The project benefited from discussions with a professor and external research, which likely contributed to its robust methodological approach.
- **Choice of TensorFlow:** Despite the tough decision, TensorFlow offers a wealth of built-in functionalities and an extensive community, making it a strong choice for complex projects.

Weaknesses:

- **Computational Constraints:** The project was complex and required high computational power, which may have limited the scope or depth of the experiments.
- **Cost:** The project incurred expenses of around \$50, primarily for GPU usage, which could be a constraint for scalability or further experiments.
- **Library Dilemma:** The decision between TensorFlow and PyTorch was challenging and could have implications for the project's ease of implementation and future adaptability.

Opportunities:

- **Scalability:** The insights gained from this project can serve as a foundation for more advanced studies that could be scaled up with increased computational resources.
- **Collaboration:** The project's success and complexity could make it an attractive candidate for academic partnerships or further external research.
- **Publication and Sharing:** Given its depth, the project holds potential for academic publication or presentations at research conferences.
- **Real-world Applications:** The models, particularly the improved ResNet-50, showed high accuracy and could be adapted for real-world applications in medical imaging, autonomous vehicles, or other fields.

Threats:

- **Technological Obsolescence:** The rapid pace of advancements in machine learning could make some aspects of the project quickly outdated.
- **Financial Constraints:** The cost associated with the project, especially if scaled up, could be prohibitive for individual researchers or smaller institutions.
- **Dependency on TensorFlow:** As the project is built on TensorFlow, any changes or updates to the library could necessitate modifications to the project.

Additional Context:

The project's complexity was accentuated by limitations in computational abilities, making decisions like the choice of libraries particularly challenging.

TensorFlow was chosen over PyTorch after significant deliberation and consultation with academic mentors, making the project a calculated risk in terms of technological choices. The project also incurred a moderate cost due to GPU requirements, adding a financial dimension to the project's challenges. These constraints make the project's successful completion and the insights gained even more noteworthy.

7 Bibliography

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8 Appendix

Code Part 1:

```
import numpy as np
import tensorflow as tf
from tensorflow.keras.layers import Conv2DTranspose, Activation
from tensorflow.keras.models import Model
import matplotlib.pyplot as plt

# Generate synthetic data
img_size = 128
num_samples = 1000
images = np.zeros((num_samples, img_size, img_size, 3), dtype=np.float32)
masks = np.zeros((num_samples, img_size, img_size, 1), dtype=np.float32)

for i in range(num_samples):
    cx, cy = np.random.randint(img_size//4, 3*img_size//4, 2)
    r = np.random.randint(img_size//8, img_size//4)
    y, x = np.ogrid[:img_size, :img_size]
    mask = (y - cy)**2 + (x - cx)**2 <= r**2
    masks[i, :, :, 0] = mask
    images[i, :, :, :] = np.random.rand(img_size, img_size, 3) * mask[...
np.newaxis]

# Define the segmentation model
input_shape = (img_size, img_size, 3)
inputs = tf.keras.Input(input_shape)
x = tf.keras.layers.Conv2D(32, (3, 3), activation='relu',
padding='same')(inputs)
x = tf.keras.layers.MaxPooling2D()(x)
x = tf.keras.layers.Conv2D(64, (3, 3), activation='relu', padding='same')(x)
x = tf.keras.layers.MaxPooling2D()(x)
x = tf.keras.layers.Conv2D(128, (3, 3), activation='relu', padding='same')(x)
x = Conv2DTranspose(64, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x)
x = Conv2DTranspose(32, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x)
x = tf.keras.layers.Conv2D(1, (3, 3), padding='same')(x)
x = Activation('sigmoid')(x)

# Compile the model
model = Model(inputs=inputs, outputs=x)
model.compile(optimizer='adam', loss='binary_crossentropy')

# Train the model
model.fit(images, masks, batch_size=32, epochs=10, validation_split=0.1)
```

```

# Get predictions
predictions = model.predict(images[:5])

# Plot images, masks, and predictions
for i in range(5):
    plt.figure(figsize=(15, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(images[i])
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(masks[i, :, :, 0], cmap='gray')
    plt.title('True Mask')
    plt.axis('off')

    plt.subplot(1, 3, 3)
    plt.imshow(predictions[i, :, :, 0], cmap='gray')
    plt.title('Predicted Mask')
    plt.axis('off')

    plt.show()

"""### Generating Synthetic Data"""

import numpy as np
import matplotlib.pyplot as plt

# Generate synthetic data
img_size = 128
num_samples = 1000
images = np.zeros((num_samples, img_size, img_size, 3), dtype=np.float32)
masks = np.zeros((num_samples, img_size, img_size, 1), dtype=np.float32)

for i in range(num_samples):
    cx, cy = np.random.randint(img_size//4, 3*img_size//4, 2)
    r = np.random.randint(img_size//8, img_size//4)
    y, x = np.ogrid[:img_size, :img_size]
    mask = (y - cy)**2 + (x - cx)**2 <= r**2
    masks[i, :, :, 0] = mask
    images[i, :, :, :] = np.random.rand(img_size, img_size, 3) * mask[...],
np.newaxis]

"""### Building Upon Baseline MobileNetV2"""

```

```

import tensorflow as tf
from tensorflow.keras.layers import Conv2D, Conv2DTranspose,
Activation
from tensorflow.keras.models import Model

# Load pre-trained MobileNetV2 model + higher level layers
base_model = tf.keras.applications.MobileNetV2(input_shape=(img_size,
img_size, 3), include_top=False, weights='imagenet')

# Create segmentation model
inputs = tf.keras.Input(shape=(img_size, img_size, 3))
x = base_model(inputs, training=False)
x = Conv2D(512, (3, 3), activation='relu', padding='same')(x)
x = Conv2DTranspose(256, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x)
x = Conv2DTranspose(128, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x)
x = Conv2DTranspose(64, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x)
x = Conv2DTranspose(32, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x)
x = Conv2DTranspose(16, (3, 3), strides=(2, 2), padding='same',
activation='relu')(x) # Additional upsampling layer
x = Conv2D(1, (3, 3), padding='same')(x)
x = Activation('sigmoid')(x)

# Compile the model
model = Model(inputs=inputs, outputs=x)
model.compile(optimizer='adam', loss='binary_crossentropy')

# Train the model
model.fit(images, masks, batch_size=32, epochs=10, validation_split=0.1)

# Get predictions
predictions = model.predict(images[:5])

# Plot images, masks, and predictions
for i in range(5):
    plt.figure(figsize=(15, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(images[i])
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(masks[i, :, :], cmap='gray')

```

```

plt.title('True Mask')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(predictions[i, :, :, 0], cmap='gray')
plt.title('Predicted Mask')
plt.axis('off')

plt.show()

"""## Image Segmentation for Oxford dataset"""

!pip install git+https://github.com/tensorflow/examples.git

import tensorflow as tf
import tensorflow_datasets as tfds
from tensorflow_examples.models.pix2pix import pix2pix
from IPython.display import clear_output
import matplotlib.pyplot as plt

"""## Load the dataset"""

dataset, info = tfds.load('oxford_iiit_pet:3.*.*', with_info=True)

def normalize(input_image, input_mask):
    input_image = tf.cast(input_image, tf.float32) / 255.0
    input_mask -= 1
    return input_image, input_mask

def load_image(datapoint):
    input_image = tf.image.resize(datapoint['image'], (128, 128))
    input_mask = tf.image.resize(datapoint['segmentation_mask'], (128, 128),
method=tf.image.ResizeMethod.NEAREST_NEIGHBOR)
    input_image, input_mask = normalize(input_image, input_mask)
    return input_image, input_mask

"""###"""

TRAIN_LENGTH = info.splits['train'].num_examples
BATCH_SIZE = 64
BUFFER_SIZE = 1000
STEPS_PER_EPOCH = TRAIN_LENGTH // BATCH_SIZE

train_images = dataset['train'].map(load_image,
num_parallel_calls=tf.data.AUTOTUNE)
test_images = dataset['test'].map(load_image,
num_parallel_calls=tf.data.AUTOTUNE)

```

```

class Augment(tf.keras.layers.Layer):
    def __init__(self, seed=42):
        super().__init__()
        self.augment_inputs = tf.keras.layers.RandomFlip(mode="horizontal",
seed=seed)
        self.augment_labels = tf.keras.layers.RandomFlip(mode="horizontal",
seed=seed)

    def call(self, inputs, labels):
        inputs = self.augment_inputs(inputs)
        labels = self.augment_labels(labels)
        return inputs, labels

train_batches = (
    train_images
    .cache()
    .shuffle(BUFFER_SIZE)
    .batch(BATCH_SIZE)
    .repeat()
    .map(Augment())
    .prefetch(buffer_size=tf.data.AUTOTUNE))

test_batches = test_images.batch(BATCH_SIZE)

def display(display_list):
    plt.figure(figsize=(15, 15))
    title = ['Input Image', 'True Mask', 'Predicted Mask']
    for i in range(len(display_list)):
        plt.subplot(1, len(display_list), i+1)
        plt.title(title[i])
        plt.imshow(tf.keras.utils.array_to_img(display_list[i]))
        plt.axis('off')
    plt.show()

base_model = tf.keras.applications.MobileNetV2(input_shape=[128, 128, 3],
include_top=False)

layer_names = [
    'block_1_expand_relu', # 64x64
    'block_3_expand_relu', # 32x32
    'block_6_expand_relu', # 16x16
    'block_13_expand_relu', # 8x8
    'block_16_project', # 4x4
]
base_model_outputs = [base_model.get_layer(name).output for name in
layer_names]

```

```

down_stack = tf.keras.Model(inputs=base_model.input,
outputs=base_model.outputs)
down_stack.trainable = False

up_stack = [
    pix2pix.upsample(512, 3), # 4x4 -> 8x8
    pix2pix.upsample(256, 3), # 8x8 -> 16x16
    pix2pix.upsample(128, 3), # 16x16 -> 32x32
    pix2pix.upsample(64, 3), # 32x32 -> 64x64
]

def unet_model(output_channels:int):
    inputs = tf.keras.layers.Input(shape=[128, 128, 3])
    skips = down_stack(inputs)
    x = skips[-1]
    skips = reversed(skips[:-1])
    for up, skip in zip(up_stack, skips):
        x = up(x)
        concat = tf.keras.layers.Concatenate()
        x = concat([x, skip])
    last = tf.keras.layers.Conv2DTranspose(filters=output_channels,
kernel_size=3, strides=2, padding='same')
    x = last(x)
    return tf.keras.Model(inputs=inputs, outputs=x)

OUTPUT_CLASSES = 3

model = unet_model(output_channels=OUTPUT_CLASSES)
model.compile(optimizer='adam',

loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
metrics=['accuracy'])

EPOCHS = 20
VAL_SUBSPLITS = 5
VALIDATION_STEPS = info.splits['test'].num_examples//BATCH_SIZE//VAL_SUBSPLITS

model_history = model.fit(train_batches, epochs=EPOCHS,
steps_per_epoch=STEPS_PER_EPOCH,
validation_steps=VALIDATION_STEPS,
validation_data=test_batches)

loss = model_history.history['loss']
val_loss = model_history.history['val_loss']

plt.figure()

```

```

plt.plot(model_history.epoch, loss, 'r', label='Training loss')
plt.plot(model_history.epoch, val_loss, 'bo', label='Validation loss')
plt.title('Training and Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss Value')
plt.ylim([0, 1])
plt.legend()
plt.show()

def create_mask(pred_mask):
    pred_mask = tf.math.argmax(pred_mask, axis=-1)
    pred_mask = pred_mask[..., tf.newaxis]
    return pred_mask[0]

def show_predictions(dataset=None, num=1):
    if dataset:
        for image, mask in dataset.take(num):
            pred_mask = model.predict(image)
            display([image[0], mask[0], create_mask(pred_mask)])
    else:
        display([sample_image,
                  create_mask(model.predict(sample_image[tf.newaxis, ...]))], sample_mask,

show_predictions(test_batches, 3)

```

Code Part 2:

```

## **Experiment 1: IRIS Image Classification**

## Load IRIS dataset
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split

iris = load_iris()
X, y = iris.data, iris.target

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

"""### Pre-Process the IRIS dataset"""

from tensorflow.keras.utils import to_categorical

# Normalize the features
X_train = X_train / X_train.max(axis=0)

```

```

X_test = X_test / X_test.max(axis=0)

# One-hot encode the labels
y_train = to_categorical(y_train, 3)
y_test = to_categorical(y_test, 3)

import tensorflow as tf

model = tf.keras.Sequential([
    tf.keras.layers.Dense(64, activation='relu', input_shape=(4,)),
    tf.keras.layers.Dense(32, activation='relu'),
    tf.keras.layers.Dense(3, activation='softmax')
])

model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history = model.fit(X_train, y_train, epochs=50, validation_data=(X_test,
y_test))

print(model.summary())

loss, accuracy = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {accuracy*100:.2f}%")

from tensorflow.keras.datasets import mnist

(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Reshape and normalize the data
X_train = X_train.reshape(-1, 28, 28, 1).astype('float32') / 255.0
X_test = X_test.reshape(-1, 28, 28, 1).astype('float32') / 255.0

from tensorflow.keras.utils import to_categorical

y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

"""# Image Classification on MNIST Dataset

## MobileNetV2
"""

from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense
from tensorflow.keras.models import Model

# Load MobileNetV2 with pre-trained weights, excluding the top layer

```



```

base_model = MobileNetV2(input_shape=(96, 96, 3), include_top=False,
weights='imagenet')

# Add custom layers
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(1024, activation='relu')(x)
predictions = Dense(10, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=predictions)

import numpy as np
from tensorflow.keras.preprocessing.image import img_to_array,
array_to_img

X_train_resized = np.array([img_to_array(array_to_img(im,
scale=False).resize((96,96)))) for im in X_train])
X_test_resized = np.array([img_to_array(array_to_img(im,
scale=False).resize((96,96)))) for im in X_test])

# Convert to RGB
X_train_resized = np.repeat(X_train_resized, 3, axis=-1)
X_test_resized = np.repeat(X_test_resized, 3, axis=-1)

model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history = model.fit(X_train_resized, y_train, epochs=5,
validation_data=(X_test_resized, y_test))

"""## Custom Model 1 for Image Classification"""

from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D,
Flatten, Dense, Dropout
from tensorflow.keras.models import Model

def CNN_Model(input_shape=(28, 28, 1), num_classes=10):
    # Define the input layer
    input_layer = Input(shape=input_shape)

    # The 1st block
    x = Conv2D(64, 3, activation='relu', padding="same")(input_layer)
    x = MaxPooling2D((2, 2), strides=2)(x)
    x = Conv2D(128, 3, padding="same", activation='relu')(x)
    x = MaxPooling2D((2, 2), strides=2)(x)

    # Flatten and add dense layers
    x = Flatten()(x)

```

```

x = Dense(256, activation='relu')(x)
x = Dropout(0.3)(x)
output_layer = Dense(num_classes, activation='softmax')(x)

# Create the Model
model = Model(inputs=input_layer, outputs=output_layer)

# Print summary of layers
print(model.summary())

return model

model = CNN_Model(input_shape=(28, 28, 1), num_classes=10)
model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history = model.fit(X_train, y_train, epochs=5, validation_data=(X_test,
y_test))

loss, accuracy = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {accuracy*100:.2f}%")

import matplotlib.pyplot as plt

# Retrieve the training and validation accuracy
train_accuracy = history.history['accuracy']
val_accuracy = history.history['val_accuracy']

# Get the number of epochs
epochs = range(1, len(train_accuracy) + 1)

# Plot the training and validation accuracy
plt.plot(epochs, train_accuracy, 'b-', label='Training Accuracy')
plt.plot(epochs, val_accuracy, 'r-', label='Validation Accuracy')
plt.title("Training and Validation Accuracy for CNN Model 1")
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()

from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D,
Flatten, Dense, Dropout, BatchNormalization
from tensorflow.keras.models import Model
from tensorflow.keras.regularizers import l1_l2

def Complex_CNN_Model(input_shape=(28, 28, 1), num_classes=10):

```

```

# Define the input layer
input_layer = Input(shape=input_shape)

# The 1st block with L1 and L2 regularization
x = Conv2D(64, 3, activation='relu', padding="same",
kernel_regularizer=l1_l2(l1=1e-5, l2=1e-4))(input_layer)
x = BatchNormalization()(x)
x = MaxPooling2D((2, 2), strides=2)(x)
x = Dropout(0.25)(x)

# The 2nd block with L1 and L2 regularization
x = Conv2D(128, 3, padding="same", activation='relu',
kernel_regularizer=l1_l2(l1=1e-5, l2=1e-4))(x)
x = BatchNormalization()(x)
x = MaxPooling2D((2, 2), strides=2)(x)
x = Dropout(0.25)(x)

# The 3rd block with L1 and L2 regularization
x = Conv2D(256, 3, padding="same", activation='relu',
kernel_regularizer=l1_l2(l1=1e-5, l2=1e-4))(x)
x = BatchNormalization()(x)
x = MaxPooling2D((2, 2), strides=2)(x)
x = Dropout(0.25)(x)

# Flatten and add dense layers
x = Flatten()(x)
x = Dense(512, activation='relu', kernel_regularizer=l1_l2(l1=1e-5, l2=1e-
4))(x)
x = Dropout(0.5)(x)
output_layer = Dense(num_classes, activation='softmax')(x)

# Create the Model
model = Model(inputs=input_layer, outputs=output_layer)

# Print summary of layers
print(model.summary())

return model

model = Complex_CNN_Model(input_shape=(28, 28, 1), num_classes=10)
model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history = model.fit(X_train, y_train, epochs=5, batch_size=128,
validation_data=(X_test, y_test)) # Increased batch size

loss, accuracy = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {accuracy*100:.2f}%")

```

```

# Retrieve the training and validation accuracy
train_accuracy = history.history['accuracy']
val_accuracy = history.history['val_accuracy']

# Get the number of epochs
epochs = range(1, len(train_accuracy) + 1)

# Plot the training and validation accuracy
plt.plot(epochs, train_accuracy, 'b-', label='Training Accuracy')
plt.plot(epochs, val_accuracy, 'r-', label='Validation Accuracy')
plt.title("Training and Validation Accuracy for CNN Model 2")
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()

```

Code Part 3:

```

import tensorflow as tf
import tensorflow_datasets as tfds

ds, info = tfds.load('malaria', split='train', shuffle_files=True,
with_info=True)

def preprocess(features):
    image = tf.image.resize(features['image'], (70, 70))
    image = tf.cast(image, tf.float32) / 255.0
    label = features['label']
    return image, label

ds = ds.map(preprocess)

train_size = int(0.8 * info.splits['train'].num_examples)
train_ds = ds.take(train_size)
val_ds = ds.skip(train_size)

train_ds = train_ds.shuffle(1024).batch(64)
val_ds = val_ds.batch(64)

from tensorflow.keras.wrappers.scikit_learn import KerasClassifier
from sklearn.model_selection import GridSearchCV
import cv2
import numpy as np
from sklearn.model_selection import train_test_split
import glob
import os

```

```

from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D,
Flatten, Dense, Dropout
from tensorflow.keras.models import Model
import matplotlib.pyplot as plt
import seaborn as sns

def create_model(optimizer='adam', dropout_rate=0.3, units=256,
learning_rate=0.001, activation='relu'):
    if optimizer == 'adam':
        optimizer = tf.keras.optimizers.Adam(learning_rate=learning_rate)
    else:
        optimizer = tf.keras.optimizers.SGD(learning_rate=learning_rate)

    input_layer = Input(shape=(70, 70, 3))
    x = Conv2D(64, 3, activation=activation, padding="same")(input_layer)
    x = MaxPooling2D((2, 2), strides=2)(x)
    x = Conv2D(128, 3, padding="same", activation=activation)(x)
    x = MaxPooling2D((2, 2), strides=2)(x)
    x = Flatten()(x)
    x = Dense(units, activation=activation)(x)
    x = Dropout(dropout_rate)(x)
    output_layer = Dense(1, activation='sigmoid')(x)
    model = Model(inputs=input_layer, outputs=output_layer)
    model.compile(optimizer=optimizer, loss="binary_crossentropy",
metrics=["accuracy"])
    return model

optimizers = ['adam', 'sgd']
dropout_rates = [0.2, 0.3, 0.4]
units_list = [128, 256, 512]
learning_rates = [0.001, 0.01]
activations = ['relu', 'tanh']

results = []
best_score = 0
best_params = {}

for optimizer in optimizers:
    for dropout_rate in dropout_rates:
        for units in units_list:
            for learning_rate in learning_rates:
                for activation in activations:
                    model = create_model(optimizer, dropout_rate, units,
learning_rate, activation)
                    model.fit(train_ds, epochs=5, validation_data=val_ds,
verbose=0)
                    _, accuracy = model.evaluate(val_ds, verbose=0)

```

```

        results.append((optimizer, dropout_rate, units, learning_rate,
activation, accuracy))
        if accuracy > best_score:
            best_score = accuracy
            best_params = {'optimizer': optimizer, 'dropout_rate':
dropout_rate,
                           'units': units, 'learning_rate': learning_rate,
'activation': activation}

    print("Best Score:", best_score)
    print("Best Params:", best_params)

    import pandas as pd

    results_df = pd.DataFrame(results, columns=['Optimizer', 'Dropout Rate',
'Units', 'Learning Rate', 'Activation', 'Accuracy'])

    import seaborn as sns
    import matplotlib.pyplot as plt

    sns.boxplot(x='Optimizer', y='Accuracy', data=results_df)
    plt.title('Effect of Optimizer on Validation Accuracy')
    plt.show()

    sns.boxplot(x='Dropout Rate', y='Accuracy', data=results_df)
    plt.title('Effect of Dropout Rate on Validation Accuracy')
    plt.show()

    sns.lineplot(x='Units', y='Accuracy', hue='Optimizer', data=results_df)
    plt.title('Effect of Units on Validation Accuracy')
    plt.show()

    sns.boxplot(x='Learning Rate', y='Accuracy', data=results_df)
    plt.title('Effect of Learning Rate on Validation Accuracy')
    plt.show()

    sns.boxplot(x='Activation', y='Accuracy', data=results_df)
    plt.title('Effect of Activation Function on Validation Accuracy')
    plt.show()

    sns.pairplot(results_df, hue="Optimizer", diag_kind="kde")
    plt.show()

    correlation_df = results_df.copy()
    correlation_df['Optimizer'] =
correlation_df['Optimizer'].astype('category').cat.codes

```

```

correlation_df['Activation']
correlation_df['Activation'].astype('category').cat.codes

correlation_matrix = correlation_df.corr()
sns.heatmap(correlation_matrix, annot=True)
plt.title('Correlation Between Hyperparameters and Accuracy')
plt.show()

# Importing required libraries
import tensorflow as tf
from tensorflow.keras import layers, Model
from tensorflow.keras.layers import Input, Conv2D, BatchNormalization,
ReLU, Add, GlobalAveragePooling2D, Dense

def residual_block(x, filters, kernel_size=3, stride=1, conv_shortcut=True):
    """
    A residual block for ResNet.
    """
    # Shortcut
    if conv_shortcut:
        shortcut = Conv2D(4 * filters, 1, strides=stride)(x)
        shortcut = BatchNormalization()(shortcut)
    else:
        shortcut = x

    # First convolution layer
    x = Conv2D(filters, 1, strides=stride)(x)
    x = BatchNormalization()(x)
    x = ReLU()(x)

    # Second convolution layer
    x = Conv2D(filters, kernel_size, padding="SAME")(x)
    x = BatchNormalization()(x)
    x = ReLU()(x)

    # Third convolution layer
    x = Conv2D(4 * filters, 1)(x)
    x = BatchNormalization()(x)

    # Adding the shortcut to the output
    x = Add()([shortcut, x])
    x = ReLU()(x)

    return x

# Improved ResNet-50 architecture
def Improved_ResNet50(input_shape=(70, 70, 3)):

```

```

"""
Build an improved version of ResNet-50.
"""

input_layer = Input(shape=input_shape)

# Initial Conv Layer
x = Conv2D(64, (7, 7), strides=(2, 2), padding="same")(input_layer)
x = BatchNormalization()(x)
x = ReLU()(x)

# Residual Blocks
x = residual_block(x, 64, stride=1, conv_shortcut=True) # First block with
conv_shortcut
x = residual_block(x, 64)
x = residual_block(x, 64)

x = residual_block(x, 128, stride=2, conv_shortcut=True) # Increasing
filters, conv_shortcut for downsampling
x = residual_block(x, 128)
x = residual_block(x, 128)
x = residual_block(x, 128)

x = residual_block(x, 256, stride=2, conv_shortcut=True) # Increasing
filters, conv_shortcut for downsampling
x = residual_block(x, 256)
x = residual_block(x, 256)
x = residual_block(x, 256)
x = residual_block(x, 256)
x = residual_block(x, 256)

x = residual_block(x, 512, stride=2, conv_shortcut=True) # Increasing
filters, conv_shortcut for downsampling
x = residual_block(x, 512)
x = residual_block(x, 512)

# Global Average Pooling layer
x = GlobalAveragePooling2D()(x)

# Fully connected layer
output_layer = Dense(1, activation='sigmoid')(x)

# Create Model
model = Model(inputs=input_layer, outputs=output_layer)

return model

# Create the improved ResNet-50 model

```



```

improved_resnet50_model = Improved_ResNet50()

results = []
best_score = 0
best_params = {}

for optimizer in optimizers:
    for dropout_rate in dropout_rates:
        for learning_rate in learning_rates:
            for activation in activations:
                model = Improved_ResNet50() # Function to create an Improved
ResNet-50 model
                # Set the optimizer, learning rate, and other hyperparameters
                model.compile(optimizer=optimizer, loss="binary_crossentropy",
metrics=["accuracy"])

                # Train the model
                model.fit(train_ds, epochs=5, validation_data=val_ds, verbose=0)

                # Evaluate the model
                _, accuracy = model.evaluate(val_ds, verbose=0)

                results.append((optimizer, dropout_rate, learning_rate, activation,
accuracy))

                if accuracy > best_score:
                    best_score = accuracy
                    best_params = {'optimizer': optimizer, 'dropout_rate':
dropout_rate,
                                'learning_rate': learning_rate, 'activation': activation}

```